

Coding Club

Let's look at python

Python's kinda neat

What's so neat about it?

1. It's easy
2. It's super versatile
3. It's used in most fields, whether they like it or not

So why do people complain?

1. It's easy
2. It's super versatile
3. It's slightly slower than other programming languages

Python / Programming basics

Input / Output

print()

Literally prints whatever is in the parenthesis
() to the console.

Will literally print anything.

```
print("Hello World!")
```

```
>>>Hello World!
```

```
print(2+2)
```

```
>>>4
```

```
print("I have " + (2+2) + " apples!")
```

```
>>>I have 4 apples!
```

input()

Will halt the program until you hit the enter / return key.

Anything you type beforehand gets used as a variable.

```
input("Press enter to continue.")
```

```
>>>Press enter to continue.
```

```
print(2+2)
input("Press enter.")
print(1+1)
```

```
>>>4
```

```
>>>Press enter.
```

```
>>>2
```

```
print(input("Type anything! "))
```

```
>>>Type anything! Hello Folks
```

```
>>>Hello Folks
```

Variables

Variables store information

Variables can hold all sorts of data like numbers, text, True/False statements, lists of data, objects, and more

Main data types

String:

- Literally just text
- Defined by quotations “”

Integer

- Plain numbers
- Positive or negative
- Just type the number

Boolean

- True / False
- No quotations “” needed
- Capitalization is important

Variable names

- A variable name must start with a letter or the underscore character
- A variable name cannot start with a number
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Variable names are case-sensitive (age, Age and AGE are three different variables)
- A variable name cannot be any of the Python keywords.

Illegal names

- 2myvariable
- my-variable
- my variable

Fix them:

- myvariable2
- my_variable
- my_variable

```
Name = "John Smith"
```

```
Age = 900
```

```
print(Name)
```

```
print(Age)
```

```
>>>John Smith
```

```
>>>900
```

```
Name = "John Smith"
```

```
Age = 900
```

```
print("My name is "+Name+" and I  
am "+Age+" years old!")
```

```
>>>My name is John Smith and I am  
900 years old!
```

```
Name = input("Enter name: ")
Age = int(input("Enter age: "))
print(name)
print(age)
```

```
>>>Enter name: John Smith
>>>Enter age: 900
>>>John Smith
>>>900
```

If - else statements

If statement

An if statement will run certain
depending on if the statement is true or
false

Else statement

If the if statement is false, run the else statement instead.

Elif statement

A hybrid of if and else. It gives you another if statement, but only if the statement(s) before it are false

```
age = 50
```

```
if age > 50:  
    print("You are not old")  
elif age < 50:  
    print("You are old")  
else:  
    print("You are 50")
```

```
>>>You are 50
```